

A Peek into Scilab

Scilab is free and open source software for numerical computation, which originated from Inria in France. It is used in various sectors like space, aeronautics, automotive, energy, defence, finance and transport. Learn more about it here.



Data analysis, in the broadest sense, requires the appropriate tools, because the possibilities offered by a spreadsheet are limited. Open source software includes a number of very interesting packages that can cover a wide range of applications. Some examples, in alphabetical order, are: GNU Octave, Python, R, Sage and Scilab. Each one of these has features of great interest. Selecting one package over another often depends on personal preferences or the need for particular aspects of Calculus. In this article, let's explore the various characteristics of Scilab via three practical examples.

Overview

Scilab (<http://www.scilab.org>) is open source software for numerical computation (Figures 1 and 2) with a syntax similar to MathWorks' Matlab. The software can be used for a wide range of applications: plotting data (2D and 3D) and for exporting to various different formats (like png, jpg, pdf, svg, etc), optimisation, statistics, signal processing,

interaction with other languages (like Python, Java, C, C++, etc), GUI building, etc. There are several additional modules available. There is also a package called Xcos (Figure 3) for modelling and simulation of systems, similar to MathWorks' Simulink. Under the CeCILL open source licence, Scilab is available free for GNU Linux, MacOS and Windows. The sources are available too. The last stable version is 5.4.1, while 5.5.0 is in beta. Scilab was created in 1990 by teams from the Institut National de Recherche en Informatique et en Automatique (Inria) and the École Nationale des Ponts et Chaussées (ENPC). Since 2012, Scilab has been developed and maintained by Scilab Enterprises.

I have chosen three practical examples to explain some of the software's Calculus and plotting capabilities—hierarchical cluster analysis, principal component analysis and the pasta design. For the first two examples, I will consider the data analysis on the quantitative determination of 17 chemical elements in 38 wines and, for the third, the study of parametric surfaces for the production of pasta. All